## LAB EXPERIMENT

## 4. Implement histogram equalization on the given image and compare it with the original image using Open CV.

## Code:

```
import cv2
import matplotlib.pyplot as plt

# Read the image
img = cv2.imread("scenary.jpg.jpg")

if img is None:
    print("Image not found. Check the file name!")
else:
    # Convert the image to grayscale
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    # Apply histogram equalization
    equalized = cv2.equalizeHist(gray)

    # Display original and equalized images
    plt.figure(figsize=(12, 8))

    plt.subplot(2, 2, 1)
    plt.imshow(gray, cmap="gray")
    plt.title("Original Grayscale Image")
    plt.axis("off")

    plt.subplot(2, 2, 2)
    plt.imshow(equalized, cmap="gray")
    plt.title("Histogram Equalized Image")
    plt.axis("off")

    # Original image histogram
    plt.subplot(2, 2, 3)
    plt.hist(gray.ravel(), 256, [0, 256])
    plt.title("Original Histogram")
    plt.xlabel("Pixel Intensity")
    plt.ylabel("Frequency")

    # Equalized image histogram
    plt.subplot(2, 2, 4)
    plt.hist(equalized.ravel(), 256, [0, 256])
    plt.title("Equalized Histogram")
    plt.xlabel("Pixel Intensity")
    plt.ylabel("Frequency")
```

```
plt.tight_layout()
plt.show()
```

